The SC-SCOPE programme – a chronicle from the second-cumulant scope to

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

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1. Purpose and scope

SC-SCOPE is the named second-cumulant-scope hypothesis of the B1 Reading-H T6 selection. The "all-orders lift" is the question whether the third-cumulant corrections preserve the selection at every certified intensity, especially the thinnest endpoint $I = 2\times10^{-3}$. This note chronicles the programme end-to-end, including a self-caught overclaim, so the logic is auditable from one place. It introduces no new result; every claim is cited.

2. The obstruction

At second-cumulant order the selection holds with STEP-5B floors $\times 59.4/\times 8.8/\times 2.6$ at $I = 4\times10^{-4}/10^{-3}/2\times10^{-3}$. The third cumulant (sunset + quartic-difference, tadpole$=0$) consumes the endpoint margin. SC-SCOPE is the hypothesis that the analysis is at matched second-cumulant order; the lift would discharge it.

3. The per-transfer arc and the honest negative

M-ENDPOINT (scscope-mendpoint-evaluation v1.1): the dressed sunset endpoint is $\times 1.13 > 1$ (the sup-grade $\times 0.97$ was a frozen-coupling artefact). **GHAT4-PERTRANSFER** (scscope-endpoint-joint-assessment v1.0): the quartic-difference shape reduction is 0.64 (convention-free, rigorous). **Joint endpoint**: the individually-positive channels jointly over-consume by $\times 1.32 \Rightarrow \times 0.757$. **Joint pairing** (scscope-joint-pairing v1.0): the most-favourable pairing recovers only $\times 0.905 < 1$; per-transfer refinements EXHAUSTED. **Scope decision** (scscope-scope-decision v1.0): second-cumulant scope ACCEPTED at the endpoint; all-orders estimate-feasible for $I \le 10^{-3}$. The named open route was a sharper STEP-5B endpoint floor $\rho \gtrsim 3.9$.

4. The floor-sharpening route – proved, with a retracted overclaim

The endpoint floor $\rho = 2.58$ used the kappa-balanced additive-energy bound $K(n_{\text{pack}}) = 8 + 4\sqrt{14}\sqrt{n_{\text{pack}}} = 103.5$, which OVERSHOOTS the Lemma-A bound at the small endpoint $n_{\text{pack}} = 40.7$. The reconciliation

$$K = \frac{\sum_{t\neq0} w_t^2}{(\lambda' I)^2} \le T'(M)$$

(R-027 $t \neq 0$ part; Parseval $w_t = \lambda' v_t$) is PROVED + verified ($K_{\text{floor}}/T' \le 0.52$), sharpening the floor to $\rho_{\text{lat}} \ge 6.55$.

Self-caught overclaim (R-029, AUDIT-2026-06-08). An initial lift used paired $= \rho/(1 + \max[R_s + R_q]) = \rho/2.872$, a LOCAL approximation at $\rho = 2.6$, concluding $\times 2.28$. The physically-correct ADDITIVE bookkeeping (the sunset is an absolute cost, the joint saturates) gives only $\times 0.945$ at $\rho_{\text{lat}} = 6.55$. The lift was RETRACTED and SC-SCOPE restored; the floor sharpening stands as a partial advance ($\times 0.757 \to \times 0.95$–$1.03$).

5. The quartic normalisation certificate

The binding question became the quartic $R_{\max} < 0.634$ (the sunset is rigorous, caps the joint at $\times 1.13$). The prior $R_{\max} = 1.019$ inherited the loose Young ceiling. The realized ratio computed directly, $R(t) = 12(5v/2)^2 \lambda'^{-2} \widehat{G^4}(t)\, 4(1 - a_0)/J(t)$, gives $R_{\max} = 0.385$. The absolute $\widehat{G^4} = (J * J)$ normalisation is PINNED by Parseval,

$$(J * J)(0)\Big|_{\text{conv}} = \frac{1}{2\pi^2} \int_0^\infty q^2 J(q)^2 \, dq, \qquad \text{ratio} = 1.0000,$$

resolving the factor-$2/(2\pi)^3$ caveat (Young consistency ratio 0.27). Hence $R_{\max} = 0.385 < 0.634$ is CERTIFIED.

6. The certified joint and its structural thinness

Under the CORRECTED additive bookkeeping,

$$\text{joint} = \frac{\text{MARGIN}}{C_2 + C_{\text{sunset}} + C_{\text{quartic}}} = \times 1.040 \ (\rho_{\text{lat}} = 6.55) \ \ldots \ \times 1.082 \ (\rho_{\text{lat}} = 12.6) > 1.$$

The certified quartic flips the prior $\times 0.945$. The closure is THIN, and the thinness is STRUCTURAL (scscope_endpoint_sweep.py): the joint SATURATES at $\times 1.13$ (sunset cap) as $\rho \to \infty$, and is sign-stable across I (the endpoint $I = 2 \times 10^{-3}$ is the thinnest; the critical $I \approx 2.5 \times 10^{-3}$ lies BEYOND it) and across $\mu^2 \in [\times 0.5, \times 2]$ (worst $\times 1.034$). A near-critical selection boundary, not a numerical artefact.

7. The lift

On re-examination the operator AUTHORISED the lift @ THIN-CERTIFIED (2026-06-09, superseding the same-day HOLD). SC-SCOPE is removed from B1's active hypothesis set,

$$\{H\text{-LAYER}, SC\text{-SCOPE}\} \longrightarrow \{H\text{-LAYER}\},$$

with B1 tier UNCHANGED T6. The all-orders endpoint selection is now certified (sunset rigorous + certified quartic), thin but sign-stable; not estimate-grade.

8. Devil's-advocate / self-adversarial review (CLAUDE.md 6.3)

(α) *"This repeats the retracted lift."* NO: the retracted lift used the local pairing $\rho/2.872$; this uses the CORRECTED additive bookkeeping and the CERTIFIED quartic. The retraction is recorded openly (AUDIT-2026-06-08).

(β) *"×1.04 is too thin."* It is thin and FLAGGED LIFTED@THIN-CERTIFIED; but every input is proved/certified and the closure is sign-stable and structural (sunset-saturating).

(γ) *"The convention could be off."* Pinned by Parseval to ratio 1.0000; a factor-2 would show as 2 or 0.5.

Quantitative sanity check. All cited: $K_{\mathrm{floor}}/T' \leq 0.52$; $\rho_{\mathrm{lat}} \geq 6.55$; Parseval ratio 1.0000; $R_{\max} = 0.385 < 0.634$; joint $\times 1.040$–$\times 1.082 > 1$; sunset saturation $\times 1.13$; μ^2-band worst $\times 1.034$.

9. Result footer

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Result ID:               B5 / SC-SCOPE programme consolidation (chronicle)
Precise statement:       Chronicle of SC-SCOPE: the second-cumulant scope, the
                         third-cumulant endpoint obstruction (sunset/quartic),
                         the per-transfer arc (M-ENDPOINT x1.13, GHAT4 0.64), the
                         joint negative (x0.757) and exhausted pairing (x0.905),
                         the floor sharpening (K<=T' proved; a retracted overclaim
                         R-029/AUDIT-2026-06-08), the quartic certificate
                         (Parseval-pinned; R_max=0.385<0.634), the certified joint
                         x1.040-x1.082>1, and the operator-enacted
                         LIFTED@THIN-CERTIFIED (B1 {H-LAYER,SC-SCOPE}->{H-LAYER}).
Scope:                   Narrative consolidation of the SC-SCOPE arc; no new claim.
Dependencies:            scscope-mendpoint-evaluation v1.1; scscope-endpoint-joint-
                         assessment v1.0; scscope-joint-pairing v1.0; scscope-scope-
                         decision v1.0; scscope-floor-sharpening v1.6 (R-029);
                         scscope-quartic-normalisation-certificate v1.0;
                         scscope_endpoint_sweep.py 4/4; R-027.
Evidence grade:          CHRONICLE (no independent result). Final state: SC-SCOPE
                         LIFTED@THIN-CERTIFIED (sunset-binding); B1 T6 on {H-LAYER}.
Reproduction command:    (none -- narrative; the cited scripts carry the asserts)
Expected output:         n/a
Falsification gate:      A more conservative sunset accounting taking the joint
                         below 1, or the Parseval ratio departing from 1; neither
                         observed.
Tier before / after:     No tier action (chronicle). Records the SC-SCOPE lift
                         and the B1 hypothesis reduction to {H-LAYER}.
No-overclaim statement:  The lift is THIN-CERTIFIED (x1.04 worst case), sunset-
                         limited near-critical -- NOT a wide-margin closure. The
                         retracted overclaim is recorded openly. B1 tier UNCHANGED
                         T6 (hypothesis reduction, not a tier lift).
Next required action:    B1 rests on H-LAYER alone; the deepest remaining piece is
                         Prop A (isotropic dressing = diagonal-Gaussian infimum).
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